Guide to Using scrcpy and ADB for Android Screen Mirroring and Interaction on Linux

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Introduction

This guide provides step-by-step instructions on using `scrcpy` and `Android Debug Bridge (ADB)` on Linux to view and control your Android device's screen on your PC. It covers both wired and wireless configurations, switching modes, and detecting connected devices.

Prerequisites

1. **Linux System Requirements:**
    - A Linux-based operating system (Ubuntu, Fedora, etc.).
    - Ensure your system has access to a terminal.
2. **Install Required Tools:**
    - **ADB:** `bash      sudo apt update      sudo apt install adb`
    - **scrcpy:** `bash      sudo apt install scrcpy`
3. **Android Device Configuration:**
    - **Enable Developer Options:**
        1. Go to **Settings > About phone**.
        2. Tap **Build number** seven times to unlock Developer Options.
    - **Enable USB Debugging:**
        1. Navigate to **Settings > System > Developer options**.
        2. Toggle on **USB debugging**.

Wired Connection Setup

1. **Connect Android Device to PC via USB Cable.**
2. **Authorize ADB Connection:**
    - On your Android device, allow USB debugging when prompted.
3. **Verify Device Detection:** `bash      adb devices`

    > You should see a list of connected devices with their unique IDs.

4. **Start scrcpy:** `bash      scrcpy`

> This launches the Android screen mirroring window.

Wireless Connection Setup

1. **Prepare Android Device and PC:**
   - Ensure your Android device and PC are on the same Wi-Fi network.
2. **Connect via USB Temporarily (First Time Only):** `bash     adb tcpip 5555`

   > This enables ADB over the network.

3. **Find Your Device's IP Address:**
   - On Android, go to **Settings > About phone > Status > IP address**.
4. **Connect Wirelessly:** `bash     adb connect <device_ip>:5555`
5. **Verify Wireless Connection:** `bash     adb devices`

   > Your device should appear in the list.

6. **Launch scrcpy:** `bash     scrcpy -e`

   > The `-e` flag specifies the most recently connected wireless device.

Switching Between Wired and Wireless Modes

- **Switch to Wired Mode:** `bash     scrcpy -d`

  > The `-d` flag selects the device connected via USB.

- **Switch to Wireless Mode:** `bash     scrcpy -e`

  > The `-e` flag selects the most recently connected wireless device.

Detecting Connected Devices

- **View Connected Devices:** `bash     adb devices`

  > Lists devices with their connection type (e.g., USB or IP).

- **Monitor Device Connections in Real-Time:** `bash watch adb devices`

  Continuously updates the list of connected devices.

Advanced Options

1. **Specify Resolution:** `bash scrcpy --max-size 1024`

   Limits the resolution to reduce latency.

2. **Record Screen:** `bash scrcpy --record file.mp4`
3. **Forward Audio (Separate Tool Required):**

   Use `sndcpy` for audio forwarding.

Troubleshooting

1. **Device Not Detected:**
   - Ensure USB debugging is enabled.
   - Try reconnecting the USB cable or restarting ADB:

   `bash adb kill-server adb start-server`
2. **Wireless Connection Issues:**
   - Ensure both devices are on the same network.
   - Verify the IP address and port.

Conclusion

Using `scrcpy` and `ADB`, you can seamlessly view and control your Android device from your Linux PC. Whether using a wired connection for stability or a wireless connection for convenience, these tools offer flexibility and powerful features to enhance your workflow.